INCH-POUND

MIL-D-70687 (AR)
13 June 1990

MILITARY SPECIFICATION

DRIVER CIRCUIT BOARD: 12594675

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the pre-amplifier assembly 12594675 herein referred to as the Driver Circuit Board, a sub-assembly of the Servo Electronics 10548193.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Material Manufacture and Inspection for General Specification
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods
MIL-STD-2000	- Standrad Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC).

10548193	- Servo Electronics
12594675	- Assembly, Driver Circuit Board

PACKAGING DATA SHEET

PS12594675	- Packaging of Driver Circuit Board

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisitions functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets for MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The materials shall be in accordance with the applicable drawings.

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3.2 Components and Assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12594675, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the driver board shall meet the requirements of MIL-F-13926 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental requirements.

3.3.1 Vibration (non-operating). Driver Circuit Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically. The cycle time shall be 120 minutes per axis. The sweep rate shall be 3.75 minutes per decade.

3.3.2 Shock. The Driver Circuit Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks, consisting of three shocks impulses of 15 $\pm$ 10% milliseconds, half-sine wave applied in each direction along each of the three mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The Driver Circuit Board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-Operating. The Driver Circuit Board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Driver Circuit Board shall meet the requirement of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

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3.3.4.2 Non-Operating (storage). The Driver Circuit Board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity Non-operating. The Driver Circuit Board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous, as specified on the schematic diagram of the Driver Circuit Board Assembly.

3.5 Performance.

3.5.1 Operation. The Driver Circuit Board shall meet the requirements of this operation while operating under test.

3.5.2 Null. Measure less than 0.33 VAC at the output terminals of the Driver Circuit Board.

3.5.3 Gain, open loop. The Driver Circuit Board shall meet the following gain requirements.

3.5.3.1 Measure between 30 MVDC and 90 MVDC at the tach input for 2.5 VAC at the output.

3.5.3.2 Measure between -30 MVDC and -90 MVDC at the tach input for 2.5 VAC at the output.

3.5.4 Gain, closed loop. The Driver Circuit Board shall meet the following requirements:

3.5.4.1 Measure between 120 VDC and 300 VDC at the tach input for 2.5 VAC at the output.

3.5.4.2 Measure between 300 VDC and 900 VDC at the DC input for 2.5 VAC at the output.

3.5.5 Current limit. The Driver Circuit Board shall meet the following requirements:

3.5.5.1 With +1 VDC input, the output shall measure between 2.5 and 3.3 VAC.

3.5.5.2 With +1 VDC input the output shall measure between 1.0 VAC and 1.6 VAC.

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3.6 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 and in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality comformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 degrees F to +90 degrees F
b.	Relative Humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Driver Circuit Boards for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I and Table II herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examination and tests.

4.4.2.1 Components and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.3.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.3.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table I shall be inspected on a 100 percent basis.

TABLE I. Classification of Defects.

	Characteristic	Requirements	Test Procedures
CRITICAL:		None	N/A
MAJOR:			
101.	Electrical	3.4	4.6.2
102.	Performance	3.5	4.6.3
MINOR:		None	N/A

4.5 Special sampling. One driver board assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL:		None	
MAJOR:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity (non-operating)	3.3.5	4.7.1.5
MINOR:		None	

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test and to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test

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equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipment listed in Table III shall be utilized in performing the applicable test of 4.6.

TABLE III. Test Equipment List.

Item Numbers	Description
1. Voltmeter, digital	0.1 mV to 100V, accuracy 0.1% ±1 digit (DC) 1.0% + 5 digit (2 to 5 KHz) (AC)
2. Oscilloscope	Input sensitivity 5 mV/Division Horizontal deflection 20 ns/Division to 5 s/Division. Dual trace. Vertical amplifier response dc to 20 MHz

4.7 Methods of inspection.

4.7.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 and as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.1.1 Vibration. The Driver Circuit Board shall be mounted on the vibration table in normal service position. The test shall be performed as specified in Method 514, Procedure VIII Category F of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration, the Driver Circuit Board shall then be tested as given in the subparagraph of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be specified in 3.3.2. After the shock impulses, the Driver Circuit shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

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4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedure I Method 504 of MIL-STD-810. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.3.4.

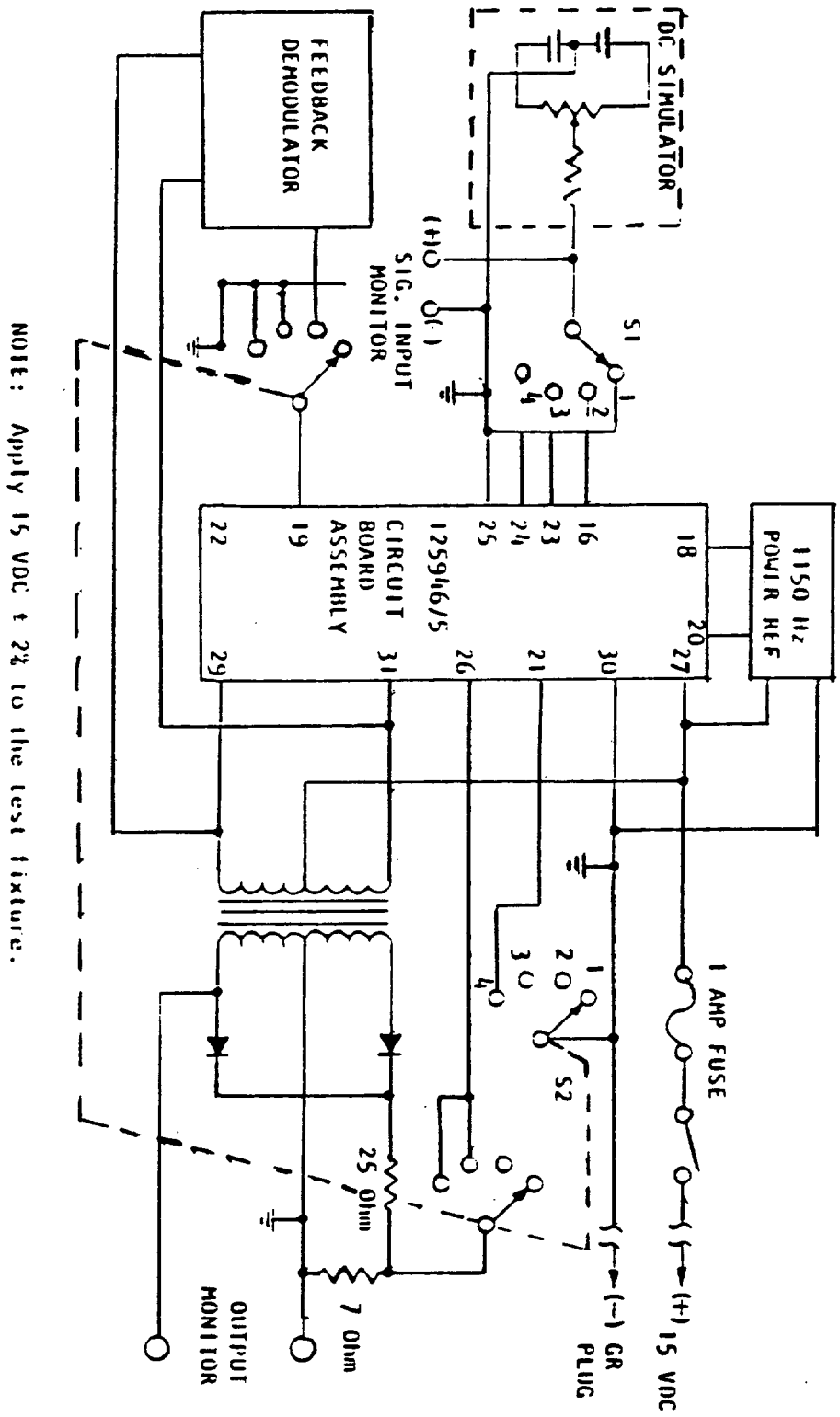
4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810 with constant temperature extremes as specified in 3.3.5. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The Driver Circuit Board shall be tested using the test set-ups as shown in Figures 1 (1.A). Apply 15 VDC $\pm$ 2% to the test fixture.

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NOTE: Apply 15 VDC ± 2% to the test fixture.

FIGURE 1

FIGURE 1

Driver Circuit Board Test Set-Up

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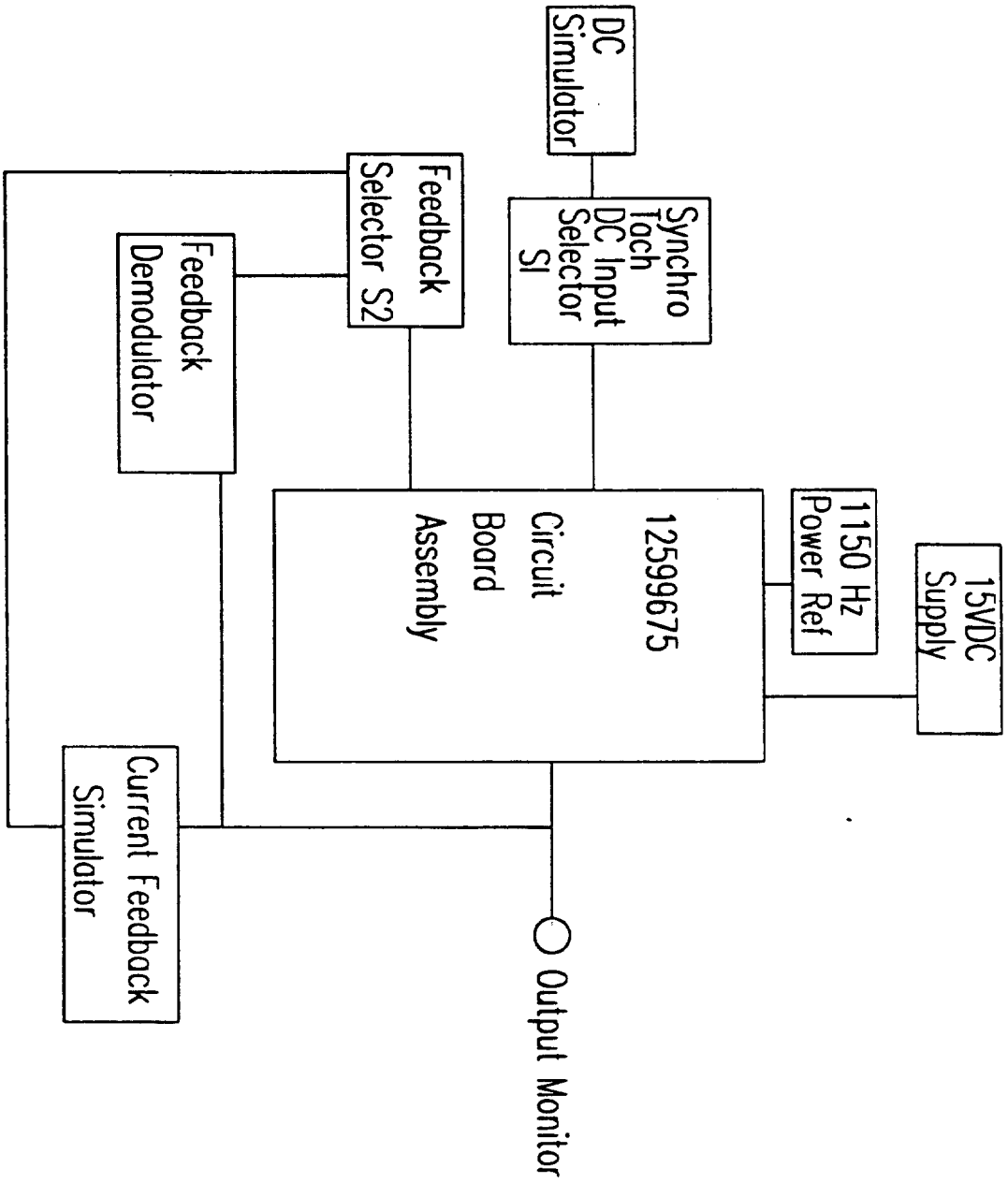


FIGURE 1A

Driver Circuit Board Set-Up (cont.)

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4.7.3.1 Operating at Null. Set S1 in position 1 and verify compliance with paragraph 3.5.2.

4.7.3.2 Gain, open loop. Set S1 in position 2 and S2 in position 1 and verify compliance with paragraph 3.5.3.

a. Verify compliance with paragraph 3.5.3.1.

b. Verify compliance with paragraph 3.5.3.2.

4.7.3.3 Gain, closed loop. Set S2 in position 2, verify compliance with paragraph 3.5.4.

a. Set S1 in position 2, verify compliance with paragraph 3.5.4.1.

b. Set S1 in position 3, verify compliance with paragraph 3.5.4.2.

c. Set S1 in position 6, verify compliance with paragraph 3.5.4.3.

4.7.3.4 Current limit. Set S1 in position 4 and verify compliance with paragraph 3.5.5.

a. Set S2 in position 3, verify compliance with paragraph 3.5.5.1.

b. Set S1 in position 4, verify paragraph 3.5.5.2.

5. PACKAGING

5.1 General. Preservation, packaging, and marking shall be in accordance with Packaging Data Sheet PS 12594675. The level of protection shall be as defined in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Driver Circuit Board is required to raise the signal to the level required for linear demodulation. It contains the tach signal input network which reduces the high frequency component of the tach signal.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the items detail specification.

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6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the items detail specification.
b. Applicable stock number.
c. First Article Sample Requirements.
d. Packaging Requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-OT-90-12009 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire Control System
Radar Set AN/VPS-2
Servo Electronics
Vulcan Air Defense System (VADS)

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1285-A768)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-70687(AR)	2. DOCUMENT DATE (YYMMDD) 13 June 1990

3. DOCUMENT TITLE
DRIVER CIRCUIT BOARD: 12594675

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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